

Due Date: 03.02

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Solution:

(a) To find the Fourier transform of $\delta(3t - 1)$, we use the formula:

$$\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\Omega t} dt$$

Substituting $u = 3t - 1$, and according to scaling property $\delta(at) = \frac{1}{|a|}\delta(t)$, we get:

$$\mathcal{F}\{\delta(3t - 1)\} = \int_{-\infty}^{\infty} \delta(u) \frac{1}{3} e^{-j\Omega \frac{u+1}{3}} du = \frac{1}{3} e^{-j\frac{\Omega}{3}}$$

Therefore, the Fourier transform of $\delta(3t - 1)$ is $\frac{1}{3}e^{-j\frac{\Omega}{3}}$.

(b) To find the Fourier transform of $\sin(\Omega_0 t + \phi_0)$, we use the formula:

$$\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\Omega t} dt$$

Expanding $\sin(\Omega_0 t + \phi_0)$ using Euler's formula, we get:

$$\sin(\Omega_0 t + \phi_0) = \frac{1}{2j}(e^{j\Omega_0 t + j\phi_0} - e^{-j\Omega_0 t - j\phi_0}) = \frac{1}{2j}(e^{j\phi_0}e^{j\Omega_0 t} - e^{-j\phi_0}e^{-j\Omega_0 t})$$

Substituting this into the Fourier transform formula and using the formula for the Fourier transform of a complex exponential, we get:

$$\mathcal{F}\{\sin(\Omega_0 t + \phi_0)\} = \frac{1}{2j}(e^{j\phi_0}2\pi\delta(\Omega - \Omega_0) - e^{-j\phi_0}2\pi\delta(\Omega + \Omega_0)) = \pi e^{j(\phi_0 - \frac{\pi}{2})}\delta(\Omega - \Omega_0) - \pi e^{-j(\phi_0 + \frac{\pi}{2})}\delta(\Omega + \Omega_0)$$

Therefore, the Fourier transform of $\sin(\Omega_0 t + \phi_0)$ is $\pi e^{j(\phi_0 - \frac{\pi}{2})}\delta(\Omega - \Omega_0) - \pi e^{-j(\phi_0 + \frac{\pi}{2})}\delta(\Omega + \Omega_0)$.

(c) To find the Fourier transform of $u(t) - u(t - T)$, we use the formula:

$$\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\Omega t} dt$$

$$\begin{aligned} \mathcal{F}\{u(t) - u(t - T)\} &= \int_{-\infty}^{\infty} [u(t) - u(t - T)]e^{-j\Omega t} dt = \int_0^T e^{-j\Omega t} dt \\ &= \left[\frac{1}{-j\Omega} e^{-j\Omega t} \right]_0^T = \frac{1}{-j\Omega} (e^{-j\Omega T} - 1) = \frac{1}{j\Omega} (1 - e^{-j\Omega T}) \end{aligned}$$

Therefore, the Fourier transform of $u(t) - u(t - T)$ is $\frac{1}{j\Omega}(1 - e^{-j\Omega T})$.

Alternative: Using the CTFT pair $u(t) \xrightarrow{\text{CTFT}} \frac{1}{j\Omega} + \pi\delta(\Omega)$, we obtain:

$$\begin{aligned}\mathcal{F}\{u(t) - u(t - T)\} &= \frac{1}{j\Omega} + \pi\delta(\Omega) - e^{j\Omega T}(\frac{1}{j\Omega} + \pi\delta(\Omega)) \\ &= \frac{1}{j\Omega}(1 - e^{-j\Omega T}) + \pi\delta(\Omega) - e^{j\Omega T}\pi\delta(\Omega) \\ &= \frac{1}{j\Omega}(1 - e^{-j\Omega T}) + \pi\delta(\Omega) - e^{j0T}\pi\delta(\Omega) \\ &= \frac{1}{j\Omega}(1 - e^{-j\Omega T})\end{aligned}$$

Grading: 18 points

- +3 proper method setup each part
- +3 correct result per part
- -1 for each minor mistake

2

Solution:

(a) By using the sampling property of delta function:

$$X(\omega) = \sum_{n=-\infty}^{\infty} \delta[n - 3]e^{-j\omega n} = e^{-j3\omega}$$

(b) According to the DTFT of a unit step function $u[n]$ that

$$u[n] \xrightarrow{\text{DTFT}} \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

Then using the *time-shifting* property of DTFT yields

$$\begin{aligned}X(\omega) &= e^{-j5\omega} \cdot \text{DTFT}\{u[n]\} = \frac{e^{-j5\omega}}{1 - e^{-j\omega}} + e^{-j5\omega}\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) \\ &= \frac{e^{-j5\omega}}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)\end{aligned}$$

where we can remove the $e^{-j5\omega}$ term on the second term since the delta function is only non-zero when $\omega = 2\pi k$, and $e^{-j5\omega}|_{\omega=2\pi k} = 1$.

Equivalently, as DTFT is assumed to be 2π -periodic, we can write the result as $\frac{e^{-j5\omega}}{1 - e^{-j\omega}} + \pi\delta(\omega)$.

(c) By definition,

$$\begin{aligned}
X(\omega) &= \sum_{n=-\infty}^{\infty} \alpha^n \sin(\omega_0 n) u[n] \cdot e^{-j\omega n} = \sum_{n=0}^{\infty} \alpha^n \cdot \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j} \cdot e^{-j\omega n} \\
&= \frac{1}{2j} \sum_{n=0}^{\infty} \left[\left(\alpha e^{-j(\omega - \omega_0)} \right)^n - \left(\alpha e^{-j(\omega + \omega_0)} \right)^n \right] \\
&= \frac{1}{2j} \cdot \left(\frac{1}{1 - \alpha e^{-j(\omega - \omega_0)}} - \frac{1}{1 - \alpha e^{-j(\omega + \omega_0)}} \right) \\
&= \frac{1}{2j} \cdot \frac{\alpha e^{-j(\omega - \omega_0)} - \alpha e^{-j(\omega + \omega_0)}}{(1 - \alpha e^{-j(\omega - \omega_0)}) (1 - \alpha e^{-j(\omega + \omega_0)})} \\
&= \frac{\alpha \sin(\omega_0) e^{-j\omega}}{(1 - \alpha e^{-j(\omega - \omega_0)}) (1 - \alpha e^{-j(\omega + \omega_0)})}
\end{aligned}$$

Alternate: Another way is to use the *multiplication property* of DTFT that

$$x_1[n] x_2[n] \xrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{2\pi} X_1(\theta) X_2(\omega - \theta) d\theta$$

Here, let us denote

$$x_1[n] = \alpha^n u[n], \quad x_2[n] = \sin(\omega_0 n).$$

The DTFT of $x_1[n]$ and $x_2[n]$ are

$$X_1(\omega) = \frac{1}{1 - \alpha e^{-j\omega}}, \quad X_2(\omega) = \frac{\pi}{j} \sum_{l=-\infty}^{\infty} [\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)]$$

Then we have

$$X(\omega) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{1 - \alpha e^{-j\theta}} \cdot \frac{\pi}{j} \sum_{l=-\infty}^{\infty} [\delta(\omega - \theta - \omega_0 - 2\pi l) - \delta(\omega - \theta + \omega_0 - 2\pi l)] d\theta$$

Here, due to the sampling property of delta function, even though we have infinite number of terms $\sum_{l=-\infty}^{\infty} \delta(\omega - \theta - \omega_0 - 2\pi l)$ to evaluate, there will only be one term $\delta(\omega - \theta - \omega_0 - 2\pi l_1)$ where $\theta = \omega - \omega_0 - 2\pi l$ is within the range $[0, 2\pi)$, and we denote the corresponding $l = l_1$, similarly, for $\delta(\omega - \theta + \omega_0 - 2\pi l)$ let us assume when $l = l_2$, $\theta = \omega + \omega_0 - 2\pi l_2$ is within $[0, 2\pi)$. Then, the DTFT above becomes

$$\begin{aligned}
X(\omega) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{1 - \alpha e^{-j\theta}} \cdot \frac{\pi}{j} [\delta(\omega - \theta - \omega_0 - 2\pi l_1) - \delta(\omega - \theta + \omega_0 - 2\pi l_2)] d\theta \\
&= \frac{1}{2j} \left[\frac{1}{1 - \alpha e^{-j(\omega - \omega_0 - 2\pi l_1)}} - \frac{1}{1 - \alpha e^{-j(\omega + \omega_0 - 2\pi l_2)}} \right] \\
&= \frac{1}{2j} \left[\frac{1}{1 - \alpha e^{-j(\omega - \omega_0)}} - \frac{1}{1 - \alpha e^{-j(\omega + \omega_0)}} \right]
\end{aligned}$$

which leads to the same result as the first approach.

(d) By using the DTFT pairs $e^{j\omega_0 n} \xrightarrow{\text{DTFT}} 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l)$, here we have $\omega_0 = \pi$, then

$$X(\omega) = 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \pi - 2\pi l)$$

Equivalently, as DTFT is assumed to be 2π -periodic, we can write the result as $2\pi\delta(\omega - \pi)$.

(e) Similar to part (c), let us denote $x_1[n] = \frac{\sin(n\pi/3)}{\pi n}$ and $x_2[n] = \frac{\sin(n\pi/6)}{\pi n}$, then according to the DTFT pair of a sinc function:

$$\frac{\sin(\omega_0 n)}{\pi n} \xrightarrow{\text{DTFT}} \begin{cases} 1, & |\omega| \leq \omega_0 \\ 0, & \omega_0 < |\omega| \leq \pi \end{cases}$$

where $|\omega_0| \leq \pi$, then the DTFT of $x_1[n]$ and $x_2[n]$ are

$$X_1(\omega) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{3} \\ 0, & \frac{\pi}{3} < |\omega| \leq \pi \end{cases}, \quad X_2(\omega) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{6} \\ 0, & \frac{\pi}{6} < |\omega| \leq \pi \end{cases}$$

To proceed, by using the *multiplication property* we obtain

$$X(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\theta) X_2(\omega - \theta) d\theta$$

One can see that the integral is a convolution between $X_1(\omega)$ and $X_2(\omega)$ on the range $[-\pi, \pi]$, then we can sketch the integral as Figure 1, where the convolution can be expressed as

$$X(\omega) = \begin{cases} \frac{1}{6}, & |\omega| \leq \frac{\pi}{6} \\ \frac{1}{4} - \frac{\omega}{2\pi}, & \frac{\pi}{6} < |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\omega| \leq \pi \end{cases}$$

Grading: 30 points

- +4 proper method setup each part
- +2 correct result per part
- -1 for each minor mistake

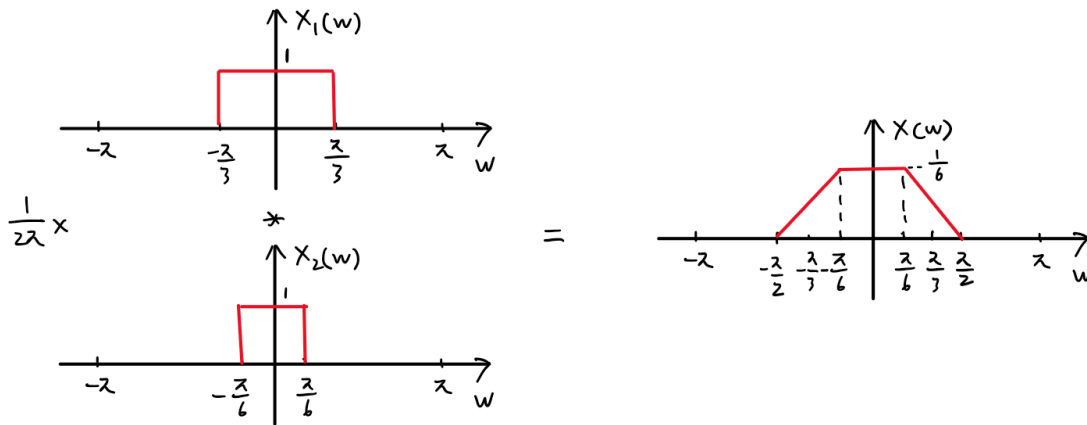


Figure 1: Problem 2(e): The sketch of $X_1(\omega)$, $X_2(\omega)$, and $X(\omega)$

Solution:

(a)

$$\begin{aligned}X_d(\omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\X_d(0) &= \sum_{n=-\infty}^{\infty} x[n] \\&= 12\end{aligned}$$

(b)

$$\begin{aligned}X_d(\omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\X_d(\pi) &= \sum_{n=-\infty}^{\infty} x[n](-1)^n \\&= -12j\end{aligned}$$

(c) Recall the form of inverse DTFT,

$$\begin{aligned}x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_d(\omega)e^{j\omega n} d\omega \\x[0] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_d(\omega) d\omega.\end{aligned}$$

Therefore, $\frac{1}{2\pi} \int_{-\pi}^{\pi} X_d(\omega) d\omega = 4\pi - j2\pi$.

(d) Start with inverse DTFT of $X_d^*(-\omega)$,

$$\begin{aligned}\frac{1}{2\pi} \int_{-\pi}^{\pi} X_d^*(-\omega)e^{j\omega n} d\omega &= -\frac{1}{2\pi} \int_{\pi}^{-\pi} X_d^*(\omega)e^{-j\omega n} d\omega \\&= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_d^*(\omega)e^{-j\omega n} d\omega \\&= \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X_d(\omega)e^{j\omega n} d\omega \right)^* \\&= x^*[n]\end{aligned}$$

The sketch would have the same real part and flipped sign of imaginary part of original $x[n]$.

Grading: 23 points

- +3 proper method setup in (a)
- +4 each proper method setup in (b)(c)(d)
- +2 each correct result
- -1 each minor mistake

Solution:

(a)

$$\begin{aligned}\sum_{n=-\infty}^{\infty} x^*[n]e^{-j\omega n} &= \left(\sum_{n=-\infty}^{\infty} x[n]e^{j\omega n} \right)^* \\ &= X_d^*(-\omega)\end{aligned}$$

(b)

$$\begin{aligned}\sum_{n=-\infty}^{\infty} x^*[-n]e^{-j\omega n} &= \left(\sum_{n=-\infty}^{\infty} x[-n]e^{j\omega n} \right)^* \\ &= \left(\sum_{m=\infty}^{-\infty} x[m]e^{-j\omega m} \right)^* \\ &= X_d^*(\omega)\end{aligned}$$

Grading: 14 points

- +7 part a
- +7 part b

5

Solution:

(a) Using the linearity and the shift property (or the pair $\delta[n - n_0] \leftrightarrow e^{-j\omega n_0}$,

$$x[n] = \delta[n - 3] + 2\delta[n - 10]$$

(b) We cannot directly apply the rule here since 3.5 here is not an integer.

$$\begin{aligned}x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j\omega 3.5} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega(n-3.5)} d\omega \\ &= \frac{1}{2\pi} \frac{e^{j\omega(n-3.5)}}{j(n-3.5)} \Big|_{-\pi}^{\pi} \\ &= \frac{1}{2\pi} \frac{e^{j\pi(n-3.5)} - e^{-j\pi(n-3.5)}}{j(n-3.5)} \\ &= \frac{1}{2\pi} \frac{2j \sin(\pi(n-3.5))}{j(n-3.5)} \\ &= \frac{(-1)^n}{\pi(n-3.5)}\end{aligned}$$

(c)

$$\begin{aligned}\cos^2(\omega) &= \frac{1 + \cos(2\omega)}{2} \\ &= \frac{1}{2} + \frac{e^{j2\omega} + e^{-j2\omega}}{4}\end{aligned}$$

$$x[n] = \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n+2] + \frac{1}{4}\delta[n-2]$$

Grading: *15 points*

- +3 proper method setup each part
- +2 correct result each part
- -1 for each minor mistake